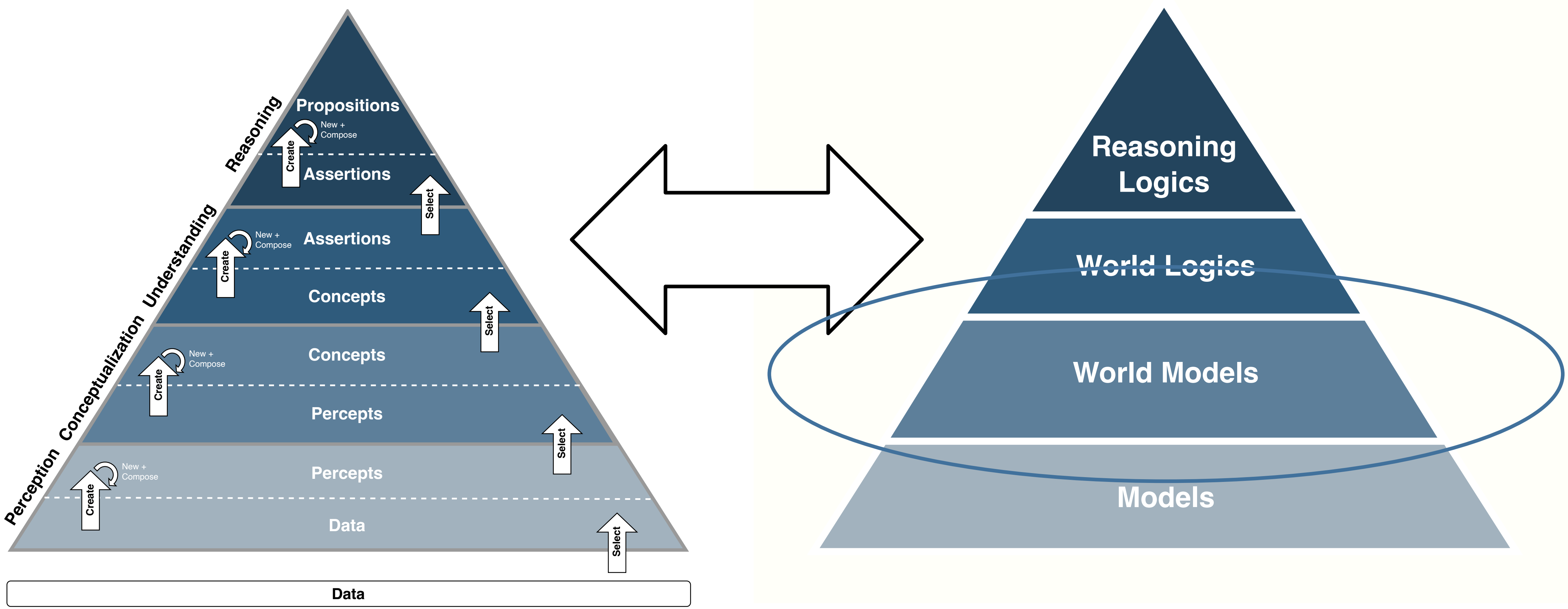
World models

(HP2T)

The knowledge Pyramid and the Logics pyramid (recall)



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World models - intuition

Intuition (World models): **Models** depict a specific instance of the world, e.g., the world under observation. **Theories** describe the contents of models. To have a complete representation of the world under observation we need to formalize the **correspondence** between theories and models.

A **World Model** is any formal notation which allows for the representation of the three types of information described above.

Observation (World models): World models do not **(!)** represent the world. They provide the means for representing the world, that is, the means for representing models, theories and how the latter represent what the former depicts.

Formal world Models – intuition

Intuition (Formal world models). **Formal world models**, also called **logical world models**, are composed of three components:

- A **domain** of interpretation which defines the boundaries within which the **intended model** can be formalized;
- A **language** describing the domain of interpretation, which defines the boundaries within which a **theory** of the intended model can be formalized;
- An **interpretation function**, that is, a functional mapping from the language to the domain of interpretation which makes explicit which elements of the language describe which elements of the domain of interpretation.

Observation (Interpretation function): A linguistic representation is always interpreted into an analogical representation. The purpose of the interpretation function is to make this mapping explicit, formally defined and unambiguous.

Types of world models - observation

Observation (Language, informal, semi-formal, Logical) There are three types of **languages** and, correspondingly, three types of **world models:**

- **Informal world models**, namely world models where the grammar of the language is defined informally, typically using Type-0 Chomsky production rules, and, as a consequence, the interpretation function cannot be and is not formally defined. Example: natural languages.
- **Semi-formal models,** namely world models where the grammar of the language is formally defined but the interpretation function is left implicit. Examples: ER models, EER models, DBs.
- **Formal (Logical) world models**, namely world models where the grammar as well as the interpretation of the language are formally defined. Example: the logics studied in this course.

Terminology (Formality of the world model). From now on, when no confusion arises, we leave implicit the type of world model (e.g., we drop the word "formal" in "formal world model").

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Interpretation function

Definition (Interpretation function) Let L_a be a language of assertions and D a domain. Then an **Interpretation Function** $\mathbf{I}_a$ is defined as

$$\mathrm{I}_a : \mathrm{L}_a \rightarrow \mathrm{D}$$

We say that a fact f ∈ M is **the interpretation of** $a \in \mathrm{I}_a$, and write

$$\mathrm{f} = \mathrm{I}_a(a) = a^{\mathrm{I}}$$

to mean that the assertion a is a linguistic description of f.

We say that f is **the interpretation of** the assertion a, or, equivalently, that a **denotes** f.

Observation (Interpretation function). Interpretation functions apply to assertions and generate facts. Different world models refine the definition of interpretation function based on the percepts they consider. See later, the definition of the interpretation functions of the different world logics. This is a key element distinguishing one logic from another.

Interpretation function (example)

- I_a(Sofia è una persona) = *Sofia ϵ person*
- I_a(Paolo è un uomo) = *Paolo ϵ man*
- I_a(Rocky is a dog) = *Rocky ϵ dog*
- I_a(Sofia is near Paolo) = *<Sofia, Paolo > ϵ near*
- I_a(Rocky è il cane di Sofia) = {*Rocky ϵ dog, <Sofia,Rocky > ϵ Owns*}
- I_a(Sofia è un’amica di Paolo) = *<Sofia, Paolo > ϵ friend*
- I_a(Sofia è bionda) = *Sofia ϵ blond*
- ….

Interpretation function – Domain of interpretation

Observation (Arbitrariety of the domain of interpretation). If one follows the sequence of previous lectures, one can notice that first we introduced the notions of domain and model, then that of language and theory, and then, as a third and last step, the notion of interpretation function. This is not by chance and it follows how things happen in practice:

- First one chooses the part of the world to be modeled, reified in the domain of interpretation;
- Second, one choose the language (alphabet and formation rules);
- Third and last, it defines the interpretation function to make sure that language means what it is meant to mean.

The domain of interpretation is the input to the overall process. Following the discussion on how things are modeled into entities (see the model theory section), the key observation is that the definition of the **domain of interpretation is arbitrary** and it depends only on the purpose of the modeler, what one wants to represent and reason about. The modeler builds the domain which is just fit to the purpose.

Interpretation function – Domain of interpretation

Observation (Arbitrariety of the domain of interpretation, examples). Some examples of domains, which have been defined in the past are:

- Entities and their properties (modeled by the LoE logic, studied later);
- Etypes and their properties;
- Complex combinations of etypes and their properties and relations (modeled by the LoD logic studied later);
- Entities, complex combinations of the etypes of the selected entities, properties and relations (modeled by the LoDe logic, studied later, as a combination of LoE and LoD);
- … and much more (see a more complete list below).

Interpretation function – non-ambiguity

Observation (Non-ambiguity) Interpretation functions, being functions, are not ambiguous. If two facts f_1 and f_2 are different it cannot be that $\mathrm{I}_a(a) = f_1$ and $\mathrm{I}_a(a) = f_2$.

Observation 1 (Polysemy) Natural language words have multiple meanings, e.g., Java, Car, bank. The polysemy of words generates the polisemy of assertions. This phenomenon, called *polysemy,* is pervasive. The average polysemy in the lexical resource *WordNet, the world de-facto* standard for lexical resources (i.e., digitized natural language vocabularies) is around 2.

Observation 2 (Polysemy). Polisemy is one of the key problems in natural language processing (NLP), with a lot of research on Word Sense Disambiguation (WSD) algorithms.

Interpretation function - synonymity

Observation (Interpretation function, synonymity). Two assertions are synonyms when they have the same meaning, that is, the interpretation of two different assertions a_1 and a_2, may denote the same fact, i.e., $\mathrm{I}_a(a_1) = \mathrm{I}_a(a_2)$. In logic synonymity is not a problem as the denotation of a word is a single percept.

Observation (Natural language, synonymity). Synonymous words are pervasive in natural languages (e.g., *car* and *automobile*).

Observation (Relational DB, synonymity). In relational DBs synonymity is not allowed, essentially for efficiency reasons (so-called *unique name assumption*). This means different strings always mean different things. That is, words behave like concepts or unique identifiers.

Interpretation function – totality and surjectivity

Observation (Totality). Interpretation functions are total. This guarantees that any element of the language has an interpretation.

Observation (Non-surjectivity). Interpretation functions are not necessarily surjective. In other words, if $I_a : L_a \rightarrow D$, L_a may not be able to name all the facts in D. This property formalizes the fact that linguistic representations do not necessarily describe all facts.

Language correctness and completeness

Definition (Language correctness and completeness). Let L $_{a}$ and D be an assertional language and a domain of interpretation, respectively. Let I_a : L_a → D be an interpretation function. Then we have two possible situations, as follows

Language correctness. Let a ∈ L_a be an assertion. If for all a, if a ∈ L_a then $I_a(a)$ =f ∈ D, then we say that L_a is **correct** with respect to D, or that D **is a domain** for L_a;

Language completeness. Let f ∈ D be a fact. If, for all f, if f ∈ D then there is an assertion a ∈ L_a such that $I_a(a)$ = f, then we say that L_a is **complete** with respect to D.

The notions of **incorrectness** and **incompleteness** are defined symmetrically.

Language correctness and completeness (continued)

Observation (Correctness of an assertional language L_a with respect to a domain D). An assertional language, to be used for a given domain, **must be correct**, that is, it must contain *only* assertions which denote facts in the reference domain. If this not the case, then we say that D is NOT a domain of L_a or, vice versa, that L_a is not a language for D. This in order to avoid nonsensical assertions (e.g., the assertion "gdhaosdf").

Observation (Completeness of an assertional language L_a with respect to a domain D). An assertional language is not necessarily complete, that is, it does not necessarily contain assertions for all the facts in a domain (which, among other things, are in principle infinite). The key feature is that it should contain all the assertions deemed relevant.

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World Model

Definition (World model). Given a **Domain of interpretation** D, a **world model** W is defined as

$$W = \langle L_a, D, I_a \rangle$$

where L_a is an **assertional language**, $I_a : L_a \rightarrow D$ is an **interpretation function** and L_a is **correct** with respect to D.

Observation (World model). L_a is not necessarily complete.

World model

$$
\begin{tikzcd}
a \arrow[d, "\mathcal{I}_a"] & \mathcal{T}_a \arrow[l, "\in"'] \arrow[r, "\subseteq"] \arrow[d, dashed, "\mathcal{I}_a"] & \mathcal{L}_a \arrow[d, "\mathcal{I}_a"] \\
\mathrm{f} & \mathrm{M} \arrow[l, "\in"'] \arrow[r, "\subseteq"] & \mathrm{D}
\end{tikzcd}
$$

Terminology – Syntax and Semantics

Terminology (Syntax and semantics). When talking about world models, people informally talk of **syntax** meaning the language of the world model, and of **semantics** meaning the domain of interpretation, associated to the syntax, via the **interpretation function,** informally or formally defined.

Observation (Syntax and semantics). Without a formal understanding of the intended semantics of a given syntax, that is, without the interpretation function and a formally defined world model, it is impossible to univocally assert whether a certain assertion (sentence) properly describes the intended model.

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World representations - intuition

Intuition (World representations): Based on the representational choices made by world models, world representations represent what is the case in the (part of the) world under consideration. They have three main components, that is:

- The **reference model** (also called the **intended model)**
- A **theory** describing the reference model
- A **correct by construction mapping**, enforced by the interpretation function which guarantees that the theory actually describes the intended model.

A **World representation** is any representation which encodes the three types of information described above.

Theory correctness and completeness

Definition (Theory correctness and completeness). Let W = ⟨L_a, D, I_a⟩ be a world model. Let $T_a \subseteq L_a$ and M ⊆ D be a set of assertions and a set of fact, respectively. Then we have the following.

Correctness. Let $a \in L_a$ be an assertion. If for all a, if $a \in T_a$ then $I_a(a) \in M$, then we say that T_a is **correct** with respect to M;

Completeness. Let f ∈ M be a fact. If, for all f, if f ∈ M then there is an assertion $a \in T_a$ such that $I_a(a) = f$, then we say that T_a is **complete** with respect to M. The completeness condition can also be written as:

If, for all $a \in L_a$, $I_a(a) \in M$ then $a \in T_a$.

The notions of **incorrectness** and **incompleteness** are defined in the obvious way.

Theory correctness and completeness (continued)

Observation (Correctness of an assertional theory T$_a$ with respect to a model M). An assertional theory T$_a$,to be used to describe an intended model, **must be correct,** that is, it must contain *only* assertions about facts in M. If this not the case then we say that M is not a model of T$_a$ or, vice versa, that T$_a$ is not a theory for M.

Observation (Completeness of an assertional theory T$_a$ with respect to a model M). An assertional theory **may be incomplete**, namely there can be facts of the model for which T$_a$ does contain assertions. Incomplete assertional theories are the default.

Observation (Correctness and completeness). The requirement on theories is the same as that on languages and domains.

World models, theories and models

Definition (Theory, model). Given a world model

$$W = \langle L_a, D, I_a \rangle$$

then, given M and T_a defined as follows,

$$M = \{f\} \subseteq D$$
$$T_a = \{a\} \subseteq L_a$$

M and T_a are, respectively, a **model** of T_a and a **theory** of M, if **T_a is correct for M.**

Reminder (correctness). T_a is correct for M, if for all a, if $a \in T_a$ then $I_a(a) \in M$

World representations

Definition (World representation). Given a world model

$$W = \langle L_a, D, I_a \rangle$$

then

$$R_W = \langle T_a, M \rangle$$

is a **world representation in** W, with

$$M = \{f\} \subseteq D$$
$$T_a = \{a\} \subseteq L_a$$

where M and T_a are respectively a **model** of T_a and a **theory** of M in W.

Observation (Theory). Theories are not necessarily complete for M.

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Maximal theories

Definition (Maximal theory). A theory is **maximal** with respect to M if it is complete with respect to M. **Not maximal** or **incomplete**, otherwise**.**

Observation 1 (Intended model, incomplete theory). The intended model has been informally defined as the model one has in mind when writing the theory. But, because of the missing requirement on completeness, an incomplete theory has multiple models, only one of which is the intended model.

Examples (Intended model, incomplete theory).

- An ER model which does not specify the relation between students and professors
- A Table describing people's job which does not specify the people's names
- The description of a photo of a dog and a person which does not have etypes "dog" and "person"

Observation 2 (Intended model, incomplete theory). The fact that the intended model is partially specified implies that the theory cannot describe all that is relevant to the problem to solve. This is the cause of hallucinations in LLMs and design mistakes in SwEng.

Canonical models

Definition (Canonical model). Given a theory, the **canonical model** for that theory is the model for which T_a is maximal.

Observation 1 (Maximal theory). Given a world model there may be mutiple maximal theories, that is theories which contain one or more assertions for each fact in the intended model. This is because of synonyms.

Observation 2 (Maximal theory). With no synonyms, there is one and only one maximal theory.

Observation 3 (Canonical model, intended model). When the theory is maximal, canonical model and intended model coincide. When this is not the case, we assume that the theory, and therefore, the canonical model are incomplete and do not fully describe what is the case.

Observation 4 (Incompletess of the canonical model). Language structures how we think about the world, but it is the world, that is, the intended model (as a mental analogic representation or as an intended model), which defines what is the case. Which is why we may have possibly incomplete /incorrect world model linguistic descriptions

Observation 5 (Canonical world representation). A world representation is **canonical** when M is the **canonical model** of T_a (that is, T_a is **maximal** for M).

Canonical World representations

Definition (World representation). Given a world model

$$W = \langle L_a, D, I_a \rangle$$

then

$$R = \langle T_a, M \rangle$$

is a **canonical world representation**, with

$$M = \{f\} \subseteq D$$
$$T_a = \{a\} \subseteq L_a$$

where M and T_a are, respectively, a **model** of T_a and a **theory** of M in W, if M is the **canonical model** of T_a (that is, T_a is **maximal** for M).

Canonical World models

Definition (Canonical world model). A world model

$$W = \langle L_a, D, I_a \rangle$$

is a **canonical world model** if

$$D = \{f\} = \{I_a(a) = f, \text{ for all } a \in L_a\}$$

That is:

$$W = \langle L_a, I_a(L_a), I_a \rangle = \langle L_a, I_a \rangle$$

Canonical World representations

Theorem (Canonical world representation). Given a canonical world model

$$W = \langle L_a, I_a \rangle$$

then

$$R = \langle T_a \rangle$$

is a **canonical world representation**, with $T_a = \{a\} \subseteq L_a$, and

$$M = \{f\} = \{I_a(a) = f, \text{ for all } a \in T_a\} \subseteq D$$

where M and T_a are, respectively, a **model** of T_a and a **theory** of M in W.

Reminder (completeness). T_a is **complete** with respect to M, If, for all $a \in L_a$, $I_a(a) \in M$ then $a \in T_a$.

Canonical world representations - observations

Observation (Canonical world representation). Once a specific world model is selected, with particular reference to the selected concepts, assertions and interpretation function, then the intended model is automatically defined once the theory (i.e., the set of assertions) is defined.

This is the standard way to use world models in practice. We have the following steps

- — Select the world model (table, graph, controlled NL)
- — Select the specific domain of interpretation and corresponding percepts, and **specify it intensionally** (e.g., as a precise mapping or function)
- — Define the interpretation function as a univocally **defined formal mapping generating the domain of interpretation**
- — Define and use world representations as canonical world representations implicitly representing the intended model and world model

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World model diversity

Observation (World diversity). The world is constantly different from itself when observed at different spatio-temporal coordinates. It changes and it evolves.

Observation (World model diversity). Assume we are describing the same world, as it appears in space and time at certain spatio-temporal coordinates. Two world models describing it may differ in the following three dimensions:

- **Domain diversity**. Two domains of the same world may differ in the percepts (types and specific instances) and facts they consider as possible. The choices of different domains is the source of the **diversity of the world**, how it is perceived by humans;
- **Language diversity**. Two languages of the same domain may differ in the alphabet and formation rules. The choice of different languages is the source of the diversity of the **descriptions of the world**, how they are provided by humans;
- **Interpretation diversity**. Two interpretation functions may differ in how alphabet elements and assertions map to percepts and vice versa. This is in fact a many-to-many mapping.

World representation diversity

Observation (World representation diversity). The notion of world representation allows to make precise the extent to which two analogical and linguistic representations may differ. Assume that two world representations share the same world model and that they describe the same (part of) the world:

- **Model diversity**. Within the bounds of the selected domain of interpretation, we may have **diversity of the facts and percepts** used to describe what is the case in the world, how it is perceived by humans;
- **Theory diversity.** Within the bounds of the selected assertional language, we may have **diversity in the alphabet** used to name percepts and **in the formation rules** used to form assertions describing the facts of the intended model. As a result we have **diversity of the assertions** used to describe the world, how they are provided by humans;
- **Interpretation diversity.** Because of the many-to-many mapping between theories and models, we may have **diversity of the theories** of the same model **and diversity of the models** of the same theory.

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Key notions

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- Semi-formal world model
- Informal world model
- Interpretation function
- Arbitrariety of domains
- Syntax and semantics
- Language correctness and completeness
- Theory correctness and completeness
- World representation
- Model
- Theory
- Maximal theory
- Canonical model
- Canonical world representation
- Canonical world model

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